KEYS EXCHANGE

Cryptography

KEY-EXCHANGE ALGORITHMS

- **Diffie-Hellman (DH)**. The Diffie-Hellman (DH) key exchange requires Alice and Bob to each agree upon a large prime number and related integer. Those two numbers can be made public, yet Alice and Bob, through mathematical computations and exchanges of intermediate values, can separately create the same key.
- **Diffie-Hellman Ephemeral (DHE)**. Whereas DH uses the same keys each time, Diffie-Hellman Ephemeral (DHE) uses different keys. ephemeral keys are temporary keys that are used only once and then discarded.
- **Elliptic Curve Diffie–Hellman (ECDH)**. Elliptic Curve Diffie–Hellman (ECDH) uses elliptic curve cryptography instead of prime numbers in its computation.
- **Perfect forward secrecy**. Public key systems that generate random public keys that are different for each session are called perfect forward secrecy. The value of perfect forward secrecy is that if the secret key is compromised, it cannot reveal the contents of more than one message.

Cryptography

KEY-EXCHANGE ALGORITHMS - DIFFIE - HELLMAN (DH)

- The question of key exchange was one of the first problems addressed by a cryptographic protocol. This was prior to the invention of public key cryptography.
- The Diffie-Hellman key agreement protocol (1976) was the first practical method for establishing a shared secret over an unsecured communication channel.
- The point is to agree on a key that two parties can use for a symmetric encryption, in such a way that an eavesdropper cannot obtain the key.

Cryptography

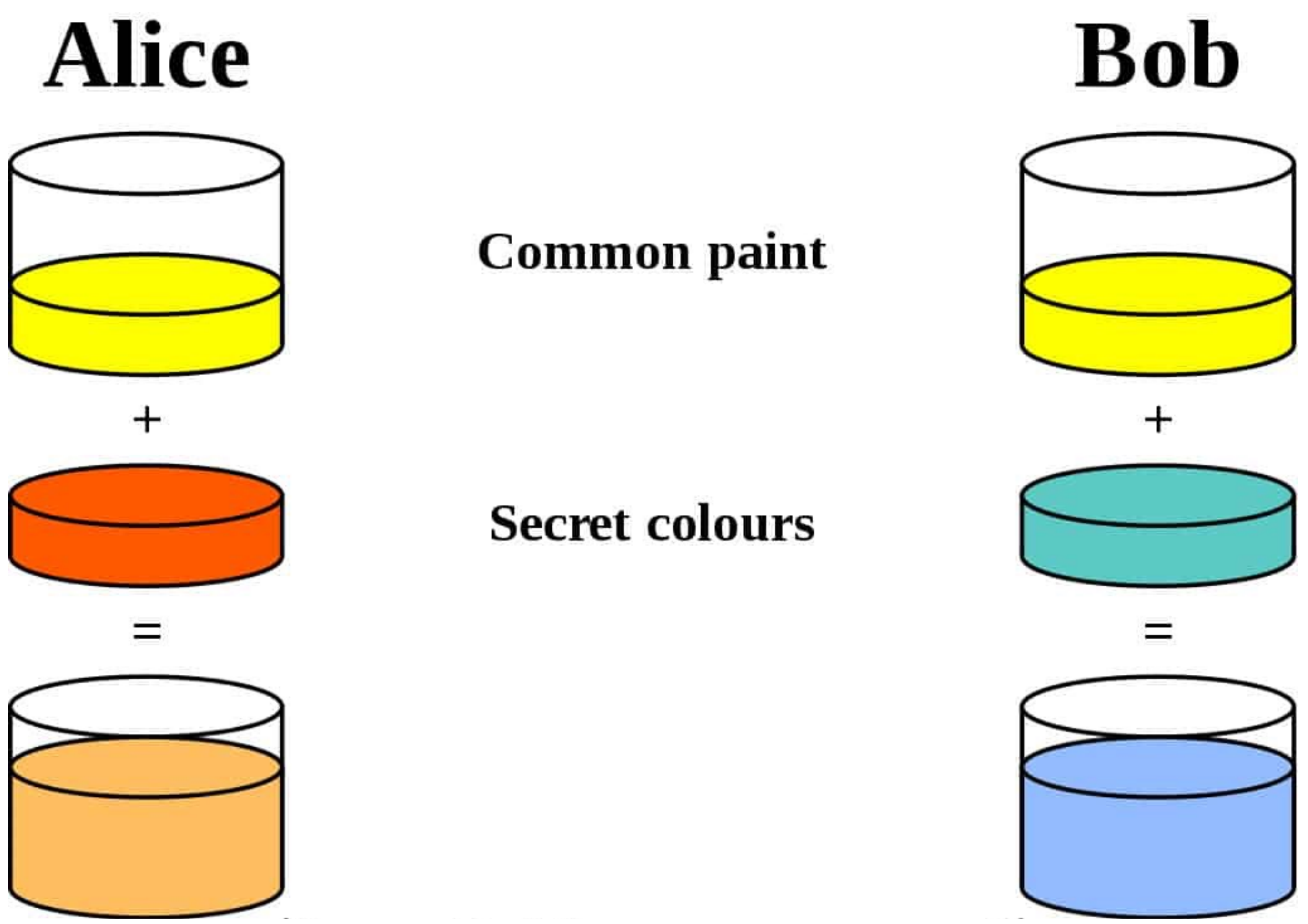
KEY-EXCHANGE ALGORITHMS - DIFFIE - HELLMAN (DH)

Cryptography

KEY-EXCHANGE ALGORITHMS - DIFFIE - HELLMAN (DH)

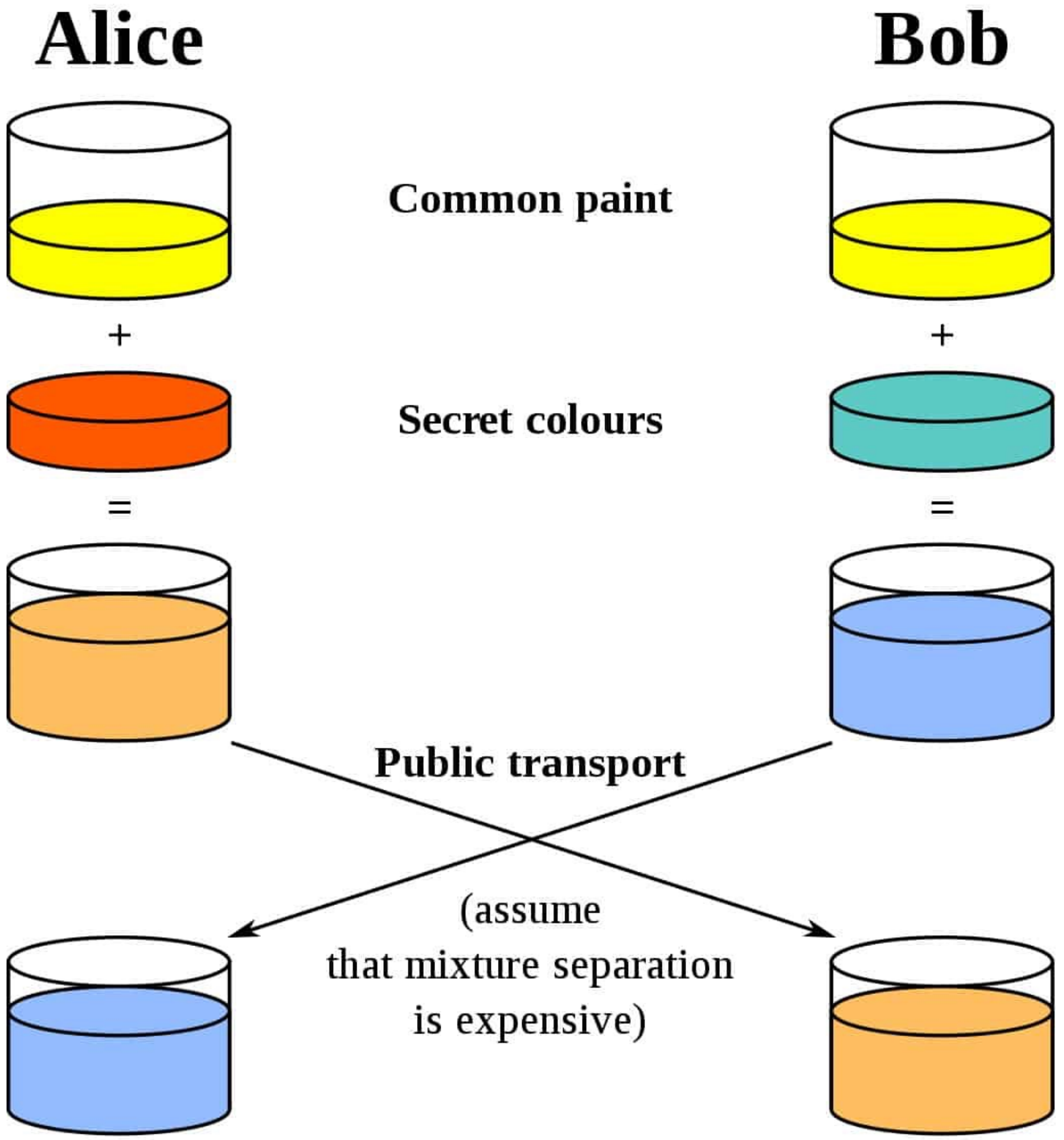
Cryptography

KEY-EXCHANGE ALGORITHMS - DIFFIE - HELLMAN (DH)



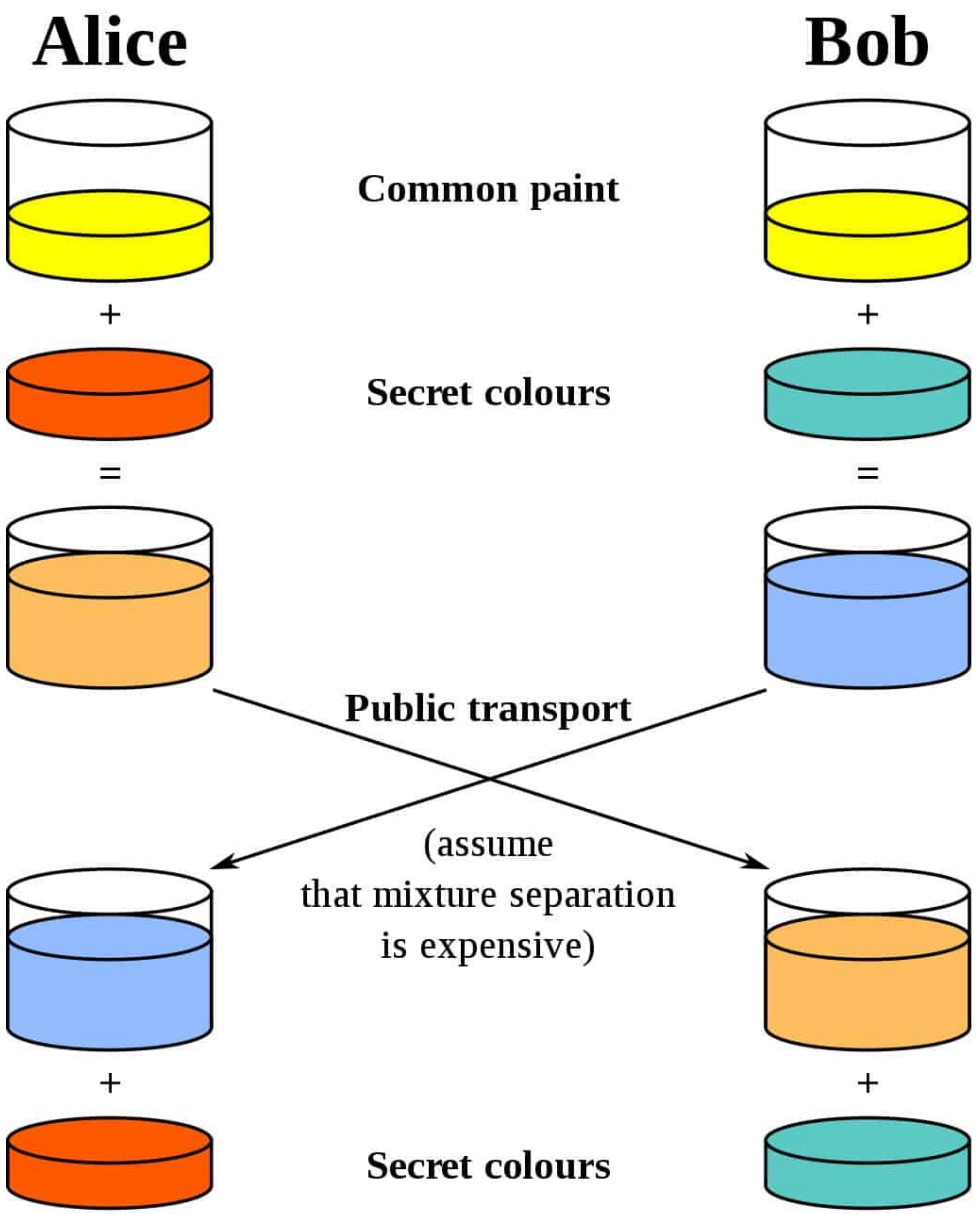
Cryptography

KEY-EXCHANGE ALGORITHMS - DIFFIE - HELLMAN (DH)



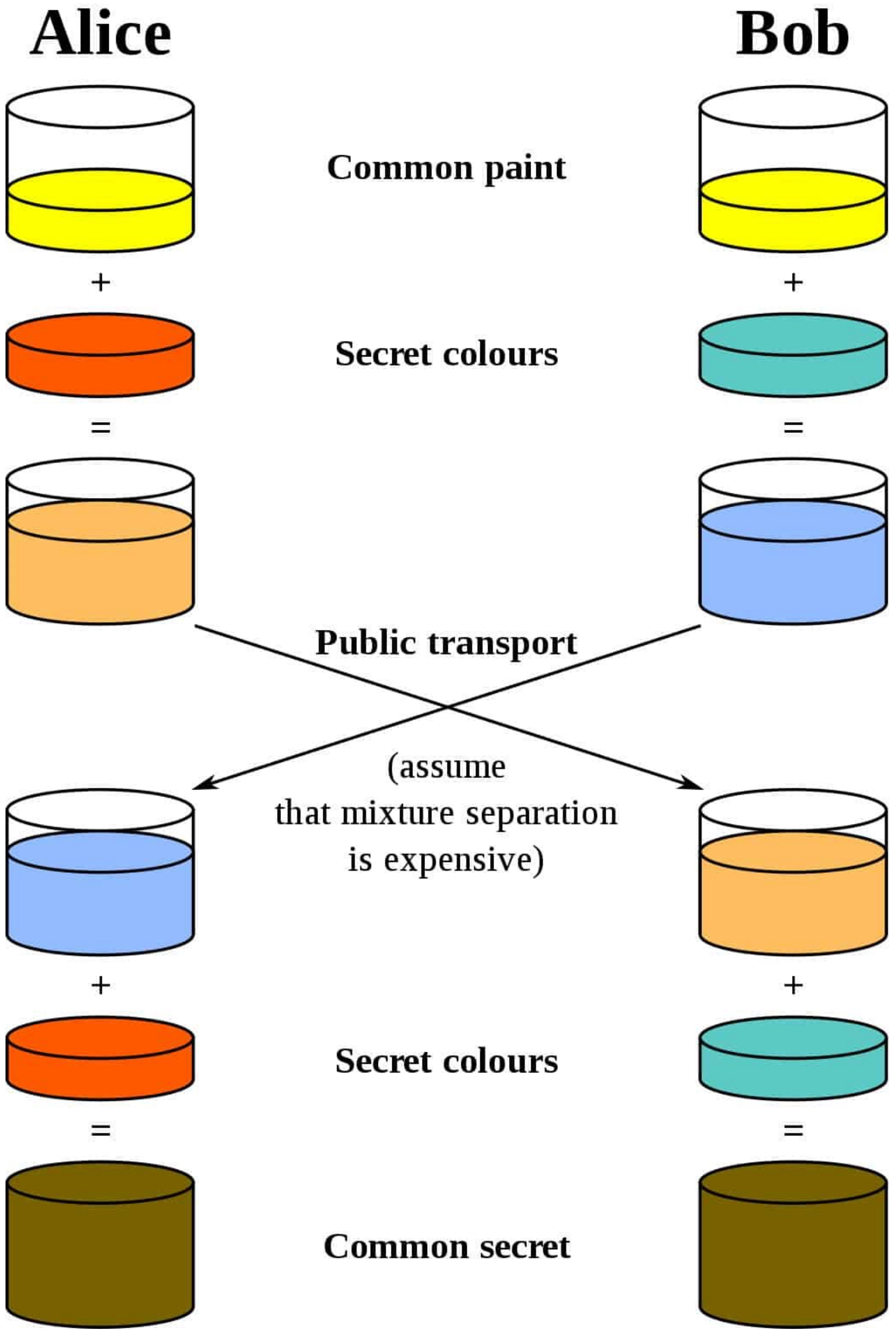
Cryptography

KEY-EXCHANGE ALGORITHMS - DIFFIE - HELLMAN (DH)



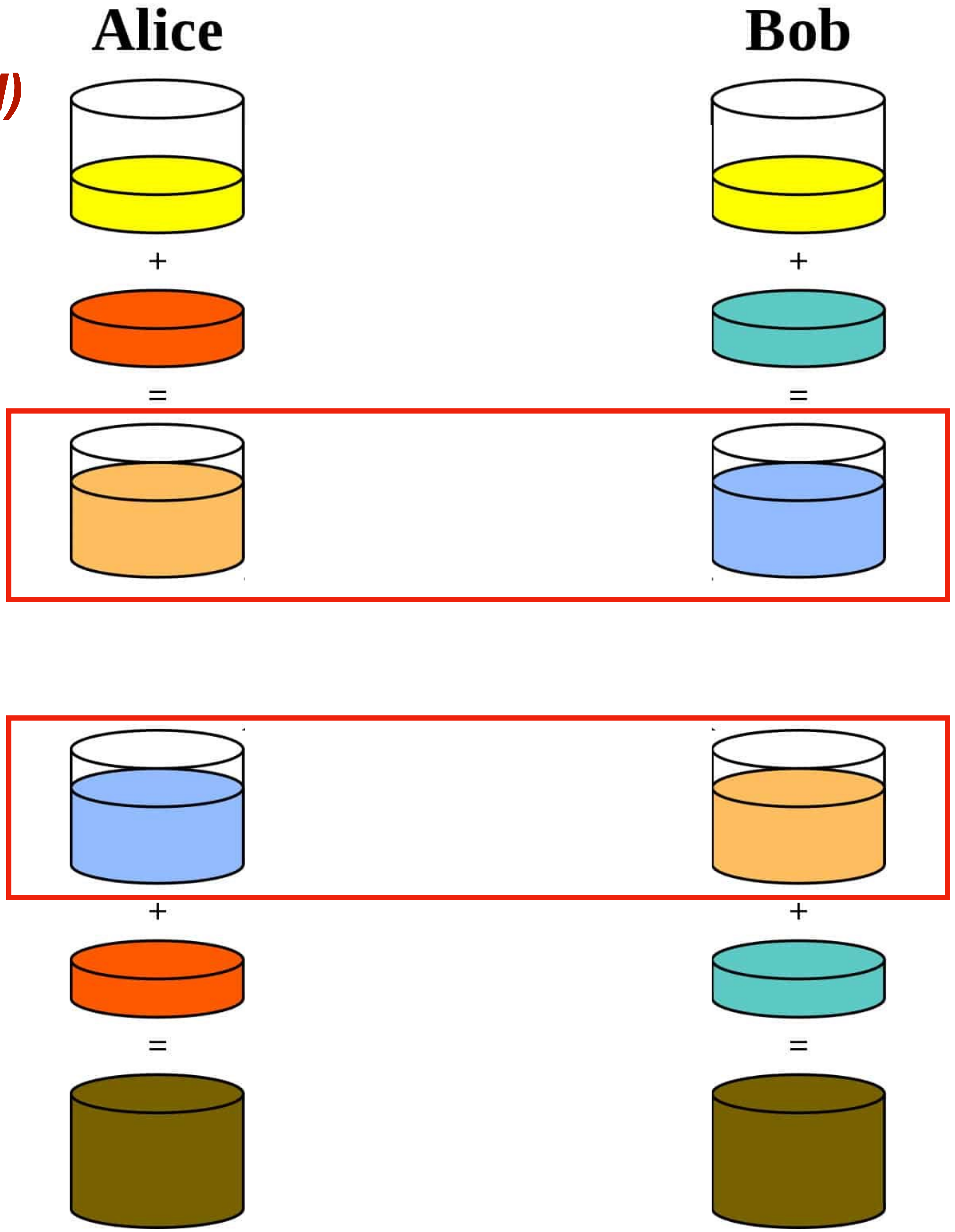
Cryptography

KEY-EXCHANGE ALGORITHMS - DIFFIE - HELLMAN (DH)



Cryptography

KEY-EXCHANGE ALGORITHMS - DIFFIE - HELLMAN (DH)



Cryptography

KEY-EXCHANGE ALGORITHMS - DIFFIE - HELLMAN (DH)

ALICE

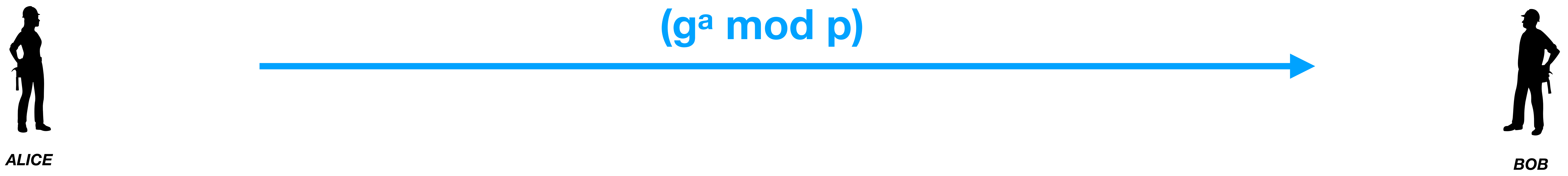
BOB

1. Alice and Bob agree on a prime number **p** and a base **g**.
2. Alice chooses a secret number **a**, and send Bob (**g^a mod p**).
3. Bob chooses a secret number **b**, and send Alice (**g^b mod p**).
4. Alice computes **((g^b mod p)^a mod p)**.
5. Bob computes **((g^a mod p)^b mod p)**.

Both Alice and Bob can use this number as their secret key.

Cryptography

KEY-EXCHANGE ALGORITHMS - DIFFIE - HELLMAN (DH)

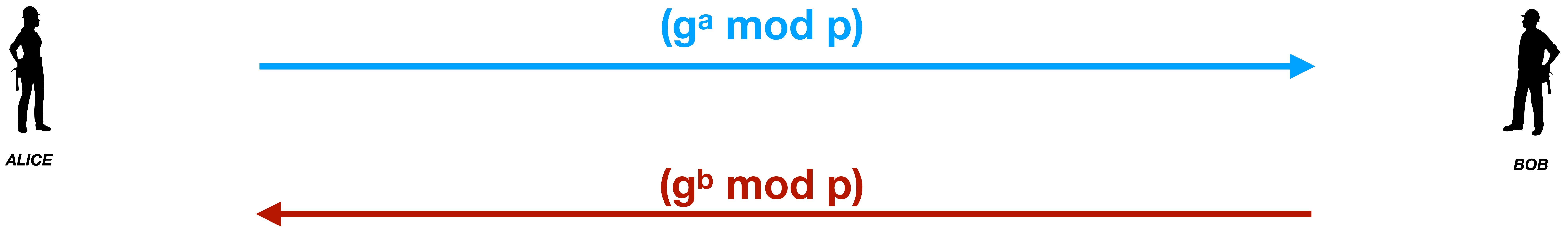


1. Alice and Bob agree on a prime number **p** and a base **g**.
2. Alice chooses a secret number **a**, and send Bob (**g^a mod p**).
3. Bob chooses a secret number **b**, and send Alice (**g^b mod p**).
4. Alice computes **((g^b mod p)^a mod p)**.
5. Bob computes **((g^a mod p)^b mod p)**.

Both Alice and Bob can use this number as their secret key.

Cryptography

KEY-EXCHANGE ALGORITHMS - DIFFIE - HELLMAN (DH)



1. Alice and Bob agree on a prime number **p** and a base **g**.
2. Alice chooses a secret number **a**, and send Bob (**g^a mod p**).
3. Bob chooses a secret number **b**, and send Alice (**g^b mod p**).
4. Alice computes **((g^b mod p)^a mod p)**.
5. Bob computes **((g^a mod p)^b mod p)**.

Both Alice and Bob can use this number as their secret key.

Cryptography

KEY-EXCHANGE ALGORITHMS - DIFFIE - HELLMAN (DH)

Why is this algorithm efficient?

Alice can compute A from the generator **g** and **a** using the "repeated squaring technique." Similarly, Alice can also compute the key ($\mathbf{g^{ab}}$) by repeated squaring technique.

What advantage does the parties have over the adversary?

Alice knows **a**, therefore she can compute $\mathbf{g^a}$ and $\mathbf{(g^b \bmod p)^a}$ efficiently. Bob knows **b**, therefore he can compute $\mathbf{g^b}$ and $\mathbf{(g^a \bmod p)^b}$ efficiently.

Adversary, however, only sees $\mathbf{g^a}$ and $\mathbf{g^b}$, and DDH states that it is computationally infeasible to distinguish $\mathbf{(g^a \bmod p)^b}$ from a random group element $\mathbf{g^r}$. Note that if the adversary can compute the discrete log $\mathbf{\log_g g^a}$, then she can easily compute $\mathbf{g^{b(\log_g g^a)}}$, the key.

Cryptography

KEY-EXCHANGE ALGORITHMS - DIFFIE - HELLMAN (DH)

ALICE → **BOB**

MALLORY (Active adversary)

ALICE

First Bob and Alice agree on a prime modulus and a generator

g mod p

g = 3 p = 17

BOB

Cryptography

KEY-EXCHANGE ALGORITHMS - DIFFIE - HELLMAN (DH)

ALICE → BOB

MALLORY (Active adversary)

ALICE

First Bob and Alice agree on a prime modulus and a generator

g mod p

g = 3 p = 17

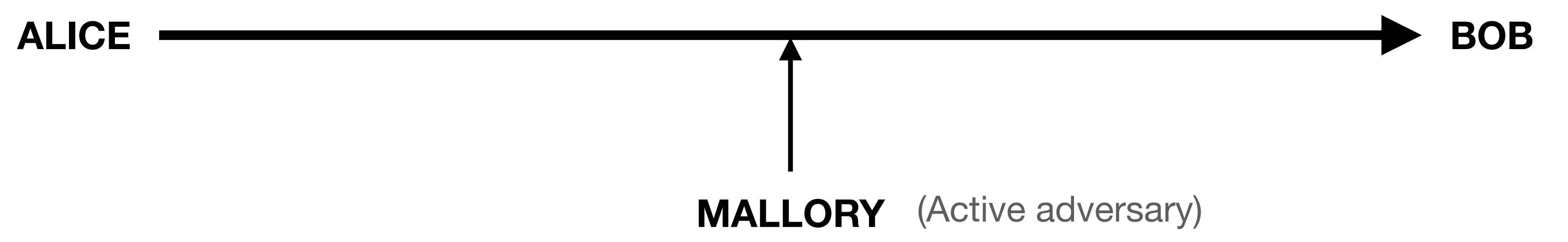
BOB

Alice then chooses a private number, lets say 15.

$$\mathbf{3^{15} \bmod 17 \equiv 6}$$

Cryptography

KEY-EXCHANGE ALGORITHMS - DIFFIE - HELLMAN (DH)



ALICE

First Bob and Alice agree on a prime modulus and a generator

g mod p

g = 3 p = 17

BOB

Alice then chooses a private number, lets say 15.

$$\mathbf{3^{15} \bmod 17 \equiv 6}$$

Send the results publicly to Bob. In this case, **6**

Mallory will help herself with the number 6 .

Cryptography

KEY-EXCHANGE ALGORITHMS - DIFFIE - HELLMAN (DH)

University of BRISTOL
Bristol Cyber Security Group

ALICE → **BOB**

MALLORY (Active adversary)


ALICE

First Bob and Alice agree on a prime modulus and a generator

g mod p

g = 3 p = 17


BOB

Alice then chooses a private number, lets say 15.

$$\mathbf{3^{15} \bmod 17 \equiv 6}$$

Send the results publicly to Bob. In this case, **6**

Mallory will help herself with the number 6 .

Bob then chooses his own random private number, lets say 13.

$$\mathbf{3^{13} \bmod 17 \equiv 12}$$

University of BRISTOL
Bristol Cyber Security Group

Cryptography

KEY-EXCHANGE ALGORITHMS - DIFFIE - HELLMAN (DH)

ALICE → BOB

↑

MALLORY (Active adversary)

ALICE

First Bob and Alice agree on a prime modulus and a generator

g mod p

←→

g = 3   p = 17

BOB

Alice then chooses a private number, lets say 15.

$$3^{15} \bmod 17 \equiv 6$$

Send the results publicly to Bob. In this case, **6**

→

Mallory will help herself with the number 6 .

Bob then chooses his own random private number, lets say 13.

$$3^{13} \bmod 17 \equiv 12$$

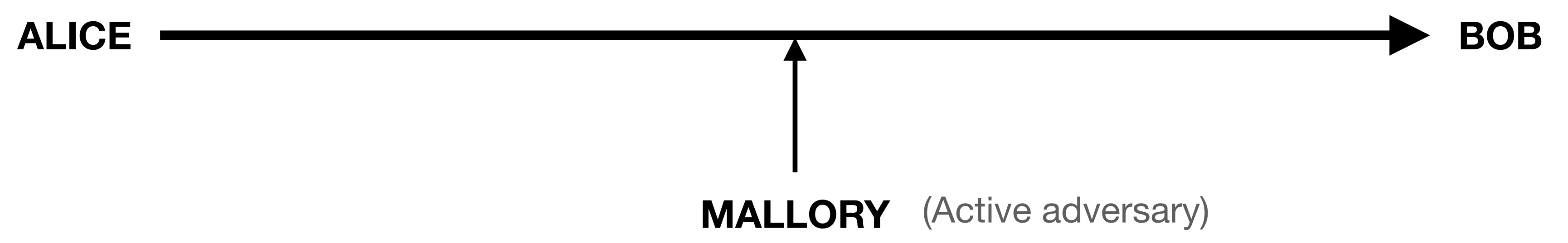
Send the results publicly to Alice. In this case, **12**

←

Mallory now has both **6** and **12**.

Cryptography

KEY-EXCHANGE ALGORITHMS - DIFFIE - HELLMAN (DH)



ALICE

First Bob and Alice agree on a prime modulus and a generator

g mod p

g = 3   p = 17

BOB

Alice then chooses a private number, lets say 15.

$$\mathbf{3^{15} \bmod 17 \equiv 6}$$

Send the results publicly to Bob. In this case, **6**

Mallory will help herself with the number 6 .

Bob then chooses his own random private number, lets say 13.

$$\mathbf{3^{13} \bmod 17 \equiv 12}$$

Send the results publicly to Alice. In this case, **12**

Mallory now has both **6** and **12**.

Alice then takes Bob's public result (**12**), and raises it to the power of her private number, **15** to obtain the private secret in this case **10**.

$$\mathbf{12^{15} \bmod 17 \equiv 10}$$

Cryptography

KEY-EXCHANGE ALGORITHMS - DIFFIE - HELLMAN (DH)

ALICE → **BOB**

MALLORY (Active adversary)

ALICE

First Bob and Alice agree on a prime modulus and a generator

g mod p

g = 3   p = 17

BOB

Alice then chooses a private number, lets say 15.

$$\mathbf{3^{15} \bmod 17 \equiv 6}$$

Send the results publicly to Bob. In this case, **6**

Mallory will help herself with the number 6 .

Bob then chooses his own random private number, lets say 13.

$$\mathbf{3^{13} \bmod 17 \equiv 12}$$

Send the results publicly to Alice. In this case, **12**

Mallory now has both **6** and **12**.

Alice then takes Bob's public result (**12**), and raises it to the power of her private number, **15** to obtain the private secret in this case **10**.

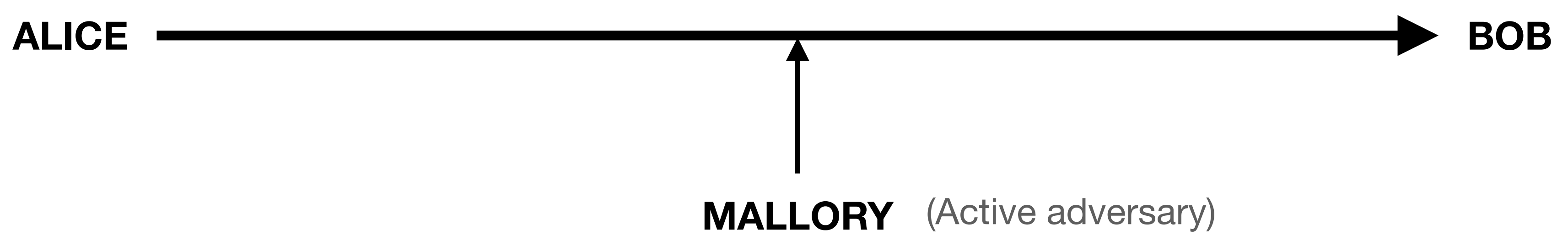
$$\mathbf{12^{15} \bmod 17 \equiv 10}$$

Bob also takes Alice's public results **6**, raises it to the power of his private number **13** to obtain the private secret in this case **10**.

$$\mathbf{6^{13} \bmod 17 \equiv 10}$$

Cryptography

KEY-EXCHANGE ALGORITHMS - DIFFIE - HELLMAN (DH)



Alice then takes Bob's public result (12), and raises it to the power of her private number, 15 to obtain the private secret in this case 10.

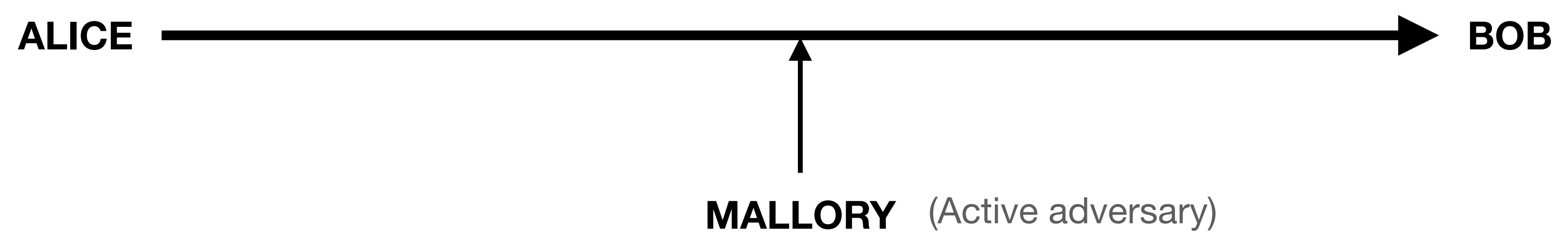
$$12^{15} \bmod 17 \equiv 10$$

Bob also takes Alice's public results 6, raises it to the power of his private number 13 to obtain the private secret in this case 10.

$$6^{13} \bmod 17 \equiv 10$$

Cryptography

KEY-EXCHANGE ALGORITHMS - DIFFIE - HELLMAN (DH)



Alice then takes Bob's public result (12), and raises it to the power of her private number, 15 to obtain the private secret in this case 10.

$$12^{15} \bmod 17 \equiv 10$$

Bob also takes Alice's public results 6, raises it to the power of his private number 13 to obtain the private secret in this case 10.

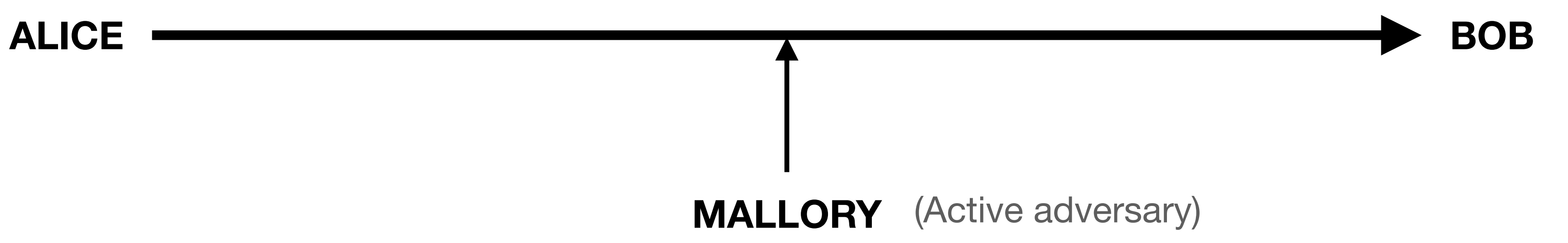
$$6^{13} \bmod 17 \equiv 10$$

$$12^{15} \bmod 17 \equiv 6^{13} \bmod 17$$

Notice this is the same calculation! It may not look like it at first but it is.

Cryptography

KEY-EXCHANGE ALGORITHMS - DIFFIE - HELLMAN (DH)



Alice then takes Bob's public result (12), and raises it to the power of her private number, 15 to obtain the private secret in this case 10.

$$12^{15} \bmod 17 \equiv 10$$

Bob also takes Alice's public results 6, raises it to the power of his private number 13 to obtain the private secret in this case 10.

$$6^{13} \bmod 17 \equiv 10$$

$$12^{15} \bmod 17 \equiv 6^{13} \bmod 17$$

Notice this is the same calculation! It may not look like it at first but it is.

Consider Alice, the 12 she got from Bob was calculated from 3^{13} mod 17 so her calculation is the same as

$${3^{13}}^{15} \bmod 17$$

Cryptography

KEY-EXCHANGE ALGORITHMS - DIFFIE - HELLMAN (DH)

Alice then takes Bob's public result (12), and raises it to the power of her private number, 15 to obtain the private secret in this case 10.

$$12^{15} \bmod 17 \equiv 10$$

Bob also takes Alice's public results 6, raises it to the power of his private number 13 to obtain the private secret in this case 10.

$$6^{13} \bmod 17 \equiv 10$$

$$12^{15} \bmod 17 \equiv 6^{13} \bmod 17$$

Notice this is the same calculation! It may not look like it at first but it is.

Consider Alice, the 12 she got from Bob was calculated from 3^{13} mod 17 so her calculation is the same as

$$3^{13^{15}} \bmod 17$$

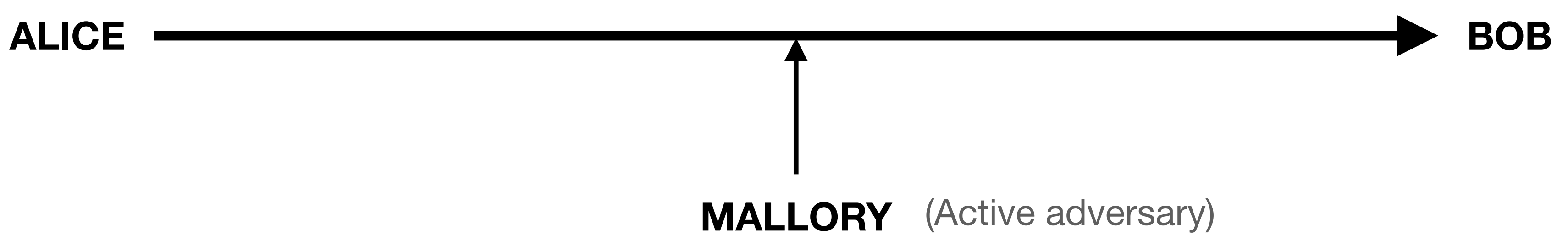
Now consider Bob, the 6 he got from Alice was calculated from 3^{15} mod 17 which is the same as

$$3^{15^{13}} \bmod 17$$

Same calculation with exponents in different order. When you flip the order of the exponents the results stays the same.

Cryptography

KEY-EXCHANGE ALGORITHMS - DIFFIE - HELLMAN (DH)



Alice then takes Bob's public result (12), and raises it to the power of her private number, 15 to obtain the private secret in this case 10.

12^{15} mod 17 $\equiv$ 10

Bob also takes Alice's public results 6, raises it to the power of his private number 13 to obtain the private secret in this case 10.

6^{13} mod 17 $\equiv$ 10

12^{15} mod 17 $\equiv$ 6^{13} mod 17

Notice this is the same calculation! It may not look like it at first but it is.

Consider Alice, the 12 she got from Bob was calculated from 3^{13} mod 17 so her calculation is the same as

${3^{13}}^{15}$ mod 17

Now consider Bob, the 6 he got from Alice was calculated from 3^{15} mod 17 which is the same as

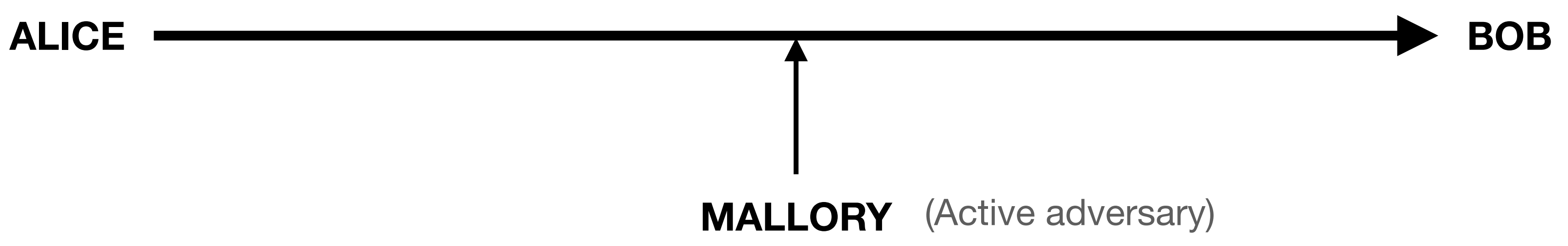
${3^{15}}^{13}$ mod 17

Same calculation with exponents in different order. When you flip the order of the exponents the results stays the same.

Mallory still has 6 and 12 but without the private numbers Mallory will never be able to find the secret number. 12^{6} mod 17 or 6^{12} mod 17

Cryptography

KEY-EXCHANGE ALGORITHMS - DIFFIE - HELLMAN (DH)



What is the shared secret?

Alice and Bob agree on prime number p = 23 and base g = 5

Alice chooses 6 to be her secret.

Bob chooses 15 to be his secret.

Alice and Bob agree on prime number p = 17 and base g = 4

Alice chooses 3 to be her secret.

Bob chooses 6 to be his secret.

Alice and Bob agree on prime number p = 17 and base g = 6

Alice chooses 4 to be her secret.

Bob chooses 5 to be his secret.